Scheduled Jumps at an FDA Decision: One Fully Worked Problem

The setup

A biotech trades at $S_0 = 100$ the session before a known FDA decision at time τ. The outcome is binary. On approval the stock jumps up by $u = 20\%$ to 120; on a complete response letter (rejection) it falls by $d = 30\%$ to 70. Under the physical measure $\mathbb{P}$ the probability of approval is $p = 0.7$. The horizon to the decision is a few days, so take the risk-free rate $r = 0$ over the window. The traded instrument is the at-the-money straddle, strike $K = 100$, expiring just after τ.

Write the jump as $S_\tau = S_0(1 + J)$ with

$$J = \begin{cases} +0.20 & \text{approval, } \mathbb{P}\text{-prob } p = 0.7, \\ -0.30 & \text{rejection, } \mathbb{P}\text{-prob } 1 - p = 0.3. \end{cases}$$

The questions (try these first)

(a) Find the risk-neutral approval probability q from no-arbitrage.

(b) Price the straddle and get the implied move $\mathbb{E}_{\mathbb{Q}}|J|$.

(c) Get the physical expected move $\mathbb{E}_{\mathbb{P}}|J|$ and the premium $\pi = \mathbb{E}_{\mathbb{Q}}|J| - \mathbb{E}_{\mathbb{P}}|J|$. Derive the closed form.

(d) Find the short-straddle P&L in each branch, and check its $\mathbb{P}$-expectation equals π.

(e) A seller has log utility and bets a fraction f of wealth. Maximize expected utility. Show $h'(f)$, solve $h'(f) = 0$, and confirm with $h''(f)$.

(f) Find the break-even round-trip transaction cost.

(g) Bonus: the mean–variance (CRRA) size for $\gamma = 3$.

Full solution

(a) Risk-neutral probability q

Under $\mathbb{Q}$ with $r = 0$, the discounted price is a martingale, so $\mathbb{E}_{\mathbb{Q}}[S_\tau] = S_0$:

$$120\,q + 70\,(1-q) = 100 \tag{1}$$
$$70 + 50\,q = 100 \tag{2}$$
$$50\,q = 30 \tag{3}$$
$$q = 0.6. \tag{4}$$

Note $q = 0.6 < p = 0.7$: the risk-neutral measure puts *less* weight on approval than the physical one. (Equivalently, under $\mathbb{P}$ the stock has positive drift, $\mathbb{E}_{\mathbb{P}}[S_\tau] = 120(0.7) + 70(0.3) = 105$, a $+5\%$ expected return; under $\mathbb{Q}$ it has none.)

(b) Straddle price and implied move

The ATM straddle pays $|S_\tau - K| = S_0\,|J|$. With $r = 0$ its price is the $\mathbb{Q}$-expected payoff:

$$\text{Straddle} = \mathbb{E}_{\mathbb{Q}}|S_\tau - 100| = q\,|120-100| + (1-q)\,|70-100| \tag{5}$$
$$= 0.6\,(20) + 0.4\,(30) = 12 + 12 = 24. \tag{6}$$

The implied move is the straddle as a fraction of spot:

$$\text{implied move} = \frac{\text{Straddle}}{S_0} = \mathbb{E}_{\mathbb{Q}}|J| = q\,u + (1-q)\,d = 0.6(0.20) + 0.4(0.30) = \boxed{0.24}. \tag{7}$$

(c) Physical move and the premium

$$\mathbb{E}_{\mathbb{P}}|J| = p\,u + (1-p)\,d = 0.7(0.20) + 0.3(0.30) = 0.14 + 0.09 = 0.23, \tag{8}$$
$$\pi = \mathbb{E}_{\mathbb{Q}}|J| - \mathbb{E}_{\mathbb{P}}|J| = 0.24 - 0.23 = \boxed{0.01}\ (1\%\text{ of spot}). \tag{9}$$

Closed form. Because $|J| = u$ on approval and $|J| = d$ on rejection,

$$\pi = \big[q\,u + (1-q)\,d\big] - \big[p\,u + (1-p)\,d\big] \tag{10}$$
$$= (q-p)\,u + \big[(1-q) - (1-p)\big]\,d \tag{11}$$
$$= (q-p)\,u + (p-q)\,d \tag{12}$$
$$= (p-q)\,(d-u). \tag{13}$$

Check: $(p-q)(d-u) = (0.7-0.6)(0.30-0.20) = (0.1)(0.1) = 0.01$. ✓

The seller profits ($\pi > 0$) exactly when the market under-weights approval ($q < p$) *and* the downside move is larger ($d > u$): the straddle is priced rich because the measure loads the bigger, worse outcome.

(d) Short-straddle P&L by branch

The seller collects 24 now and pays $|S_\tau - 100|$ at τ (per 100 of spot, i.e. in % of spot):

$$\text{approval:} \quad +24 - 20 = +4 \ \Rightarrow\ g_+ = +0.04, \tag{14}$$
$$\text{rejection:} \quad +24 - 30 = -6 \ \Rightarrow\ g_- = -0.06. \tag{15}$$

Expected P&L under $\mathbb{P}$:

$$\pi_s = p\,g_+ + (1-p)\,g_- = 0.7(0.04) + 0.3(-0.06) = 0.028 - 0.018 = 0.01 = \pi. \checkmark \tag{16}$$

(e) Optimal size under log utility (every derivative)

Stake a fraction f of wealth; terminal wealth is $W_0(1+f\tilde{R})$ with $\tilde{R} = g_+$ w.p. 0.7 and g_- w.p. 0.3. Maximize

$$h(f) = \mathbb{E}_{\mathbb{P}}\big[\ln(1+f\tilde{R})\big] = 0.7\,\ln(1+0.04f) + 0.3\,\ln(1-0.06f). \tag{17}$$

First derivative (chain rule, $\frac{d}{dx}\ln(1+ax) = \frac{a}{1+ax}$):

$$h'(f) = 0.7\cdot\frac{0.04}{1+0.04f} + 0.3\cdot\frac{-0.06}{1-0.06f} = \frac{0.028}{1+0.04f} - \frac{0.018}{1-0.06f}. \tag{18}$$

Set $h'(f) = 0$:

$$\frac{0.028}{1+0.04f} = \frac{0.018}{1-0.06f} \tag{19}$$

$$0.028\,(1-0.06f) = 0.018\,(1+0.04f) \tag{20}$$

$$0.028 - 0.00168f = 0.018 + 0.00072f \tag{21}$$

$$0.028 - 0.018 = 0.00072f + 0.00168f \tag{22}$$

$$0.010 = 0.0024\,f \tag{23}$$

$$f^\star = \frac{0.010}{0.0024} = \boxed{4.1\overline{6}}. \tag{24}$$

Check with the closed form $f^\star = -\,\pi_s/(g_+ g_-)$:

$$f^\star = \frac{-\,0.010}{(0.04)(-0.06)} = \frac{-0.010}{-0.0024} = 4.1\overline{6}.\checkmark \tag{25}$$

Second derivative confirms a maximum (every term negative):

$$h''(f) = -\,0.7\cdot\frac{(0.04)^2}{(1+0.04f)^2} - 0.3\cdot\frac{(0.06)^2}{(1-0.06f)^2} < 0 \quad \text{for all admissible } f. \tag{26}$$

So h is strictly concave and $f^\star$ is the unique maximizer. (The size 4.17× wealth is a Kelly *overbet*; it is huge because this two-point bet has a tiny loss, $g_- = -6\%$, and no fat tail. The paper's caveat applies: size to the worst plausible gap, not to this toy variance.)

(f) Break-even transaction cost

Let κ be the round-trip bid-ask spread as a fraction of the straddle price $c = \mathbb{E}_{\mathbb{Q}}|J| = 0.24$. The seller pays κc, so net expected P&L is $\pi_s - \kappa c$, positive iff

$$\kappa < \kappa^\star = \frac{\pi_s}{c} = \frac{0.01}{0.24} = \boxed{0.041\overline{6}} \approx 4.2\% \text{ of the straddle.} \tag{27}$$

(g) Bonus: mean–variance (CRRA) size, $\gamma = 3$

Variance of the P&L:

$$\mathbb{E}_{\mathbb{P}}[\tilde{R}^2] = 0.7(0.04)^2 + 0.3(0.06)^2 = 0.7(0.0016) + 0.3(0.0036) = 0.00112 + 0.00108 = 0.00220, \tag{28}$$

$$\sigma^2 = \mathbb{E}[\tilde{R}^2] - (\mathbb{E}[\tilde{R}])^2 = 0.00220 - (0.01)^2 = 0.00220 - 0.00010 = 0.00210. \tag{29}$$

The CRRA approximation $f^\star \approx \pi_s/(\gamma\sigma^2)$ gives, for $\gamma = 3$,

$$f^\star \approx \frac{0.01}{3\,(0.00210)} = \frac{0.01}{0.0063} \approx 1.59. \tag{30}$$

(For $\gamma = 1$ it gives $0.01/0.0021 \approx 4.76$, near the exact log value 4.17; the small gap is because the approximation uses σ^2 while the exact log FOC uses $\mathbb{E}[\tilde{R}^2]$.)

What each part exercised

(a) change of measure / no-arbitrage. (b) risk-neutral pricing as an expectation. (c) the premium and its closed form $\pi = (p - q)(d - u)$. (d) the trade's payoff and why its mean is the premium. (e) constrained optimization with the full first- and second-order conditions and the Kelly closed form. (f) the cost threshold. (g) the mean–variance size.